RAJALAKSHMI ENGINEERING COLLEGE

ABINAVVA 251001504-2025-IT

A2

Started on	Tuesday, 22 September 2026, 8:55 AM
State	Finished
Completed on	Friday, 9 October 2026, 11:09 AM
Time taken	17 days 2 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, count = 0;
    scanf("%d", &n);

    int c = 0;
    count++;                        // c = 0

    for(int i = n/2; i < n; i++)
    {
        count++;                    // i < n

        for(int j = 1; j < n; j = 2*j)
        {
            count++;                // j < n

            for(int k = 1; k < n; k = k*2)
            {
                count++;            // k < n

                c++;
                count++;            // c++
            }

            count++;                // final k < n
        }

        count++;                    // final j < n
    }

    count++;                        // final i < n

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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